Arcsecond.local on your network

Arcsecond.local listens on one port of your local network (5555) and on nothing from the Internet side. Everything else is outbound. One destination is required: licensing.arcsecond.io on port 443. A default installation also looks up 10 public astronomy services, all on port 443, and runs without them. The rest exists only when you turn a feature on.

WHAT LISTENS

- Port 5555, every interface: The application itself.
- Port 8800, this machine only: The programming interface, for the arcsecond command and scripts run on the machine itself.
- Port 8765, every interface: Relays webcam and all-sky camera images to the installation. Runs only when cameras are registered. Optional.

WHAT IT CONTACTS

Required. Outbound, encrypted.

`licensing.arcsecond.io`	443	Licence activation and daily check

Public catalogues and lookups. Outbound, encrypted. The installation runs without them, with less to show.

`exoplanetarchive.ipac.caltech.edu`	443	Exoplanets and transits (NASA archive)
`ogle.astrouw.edu.pl`	443	Microlensing candidates (OGLE)
`minorplanetcenter.net, cgi.minorplanetcenter.net`	443	Near-Earth object candidates
`celestrak.org`	443	Satellite orbits
`simbad.cds.unistra.fr`	443	Object names and coordinates (SIMBAD)
`timezonedb.com`	443	Time-zone table refresh
`statics.arcsecond.io`	443	Sky darkness of a site
`ssd.jpl.nasa.gov, ssd-api.jpl.nasa.gov`	443	Comet and asteroid positions
`skyview.gsfc.nasa.gov`	443	Finding charts (NASA SkyView)
`datacenter.iers.org, maia.usno.navy.mil, hpiers.obspm.fr, data.iana.org`	443	Earth rotation tables

Only when you turn the feature on. Outbound. Nothing here is contacted by a default installation.

`scixplorer.org`	443	Papers about a target
`auth.gcn.nasa.gov`	443	Transient alerts: sign-in
`kafka, kafka1, kafka2, kafka3 .gcn.nasa.gov`	9092	Transient alerts: the feed
`kafka-ztf.fink-broker.org`	24499, not encrypted	Survey alerts (Fink)
`api.tomorrow.io`	443	Weather forecast
`www.wis-tns.org`	443	Transient Name Server lookups
`a server you choose`	445, 22, 21 or 443	Copies to a storage you attach
`a server you choose`	587, 465 or 25	Email through your mail server

To install and to update. Outbound, encrypted. Not needed while running.

`ghcr.io, pkg-containers.githubusercontent.com`	443	Arcsecond.local itself
`registry-1.docker.io, auth.docker.io, production.cloudflare.docker.com`	443	Database and message broker
`pypi.org, files.pythonhosted.org`	443	The arcsecond command

From your users' browsers. Outbound, encrypted, from their workstations, not from the installation.

`server.arcgisonline.com, tile.openstreetmap.org, global.ssl.fastly.net, api.maptiler.com`	443	Map backgrounds
`alasky.cds.unistra.fr, alaskybis.cds.unistra.fr, alasky.unistra.fr, skyview.gsfc.nasa.gov`	443	Sky survey pictures
`statics.arcsecond.io`	443	Light-pollution layer, illustrations

On your local network only. Never leaves the site.

`addresses you enter`	set per device	Telescopes, domes, cameras, weather
`a broadcast on your network`	32227 (datagram)	Finding instruments on the network
`addresses you enter`	554, 80 or 443	Webcams and all-sky cameras

WHAT NEVER LEAVES THE SITE

- Images and data files
- Night logs, sessions and observing conditions
- Equipment configuration and instrument state
- Accounts, passwords and access keys of the installation's users
- Usage statistics, error reports or any other telemetry

IF THE CONNECTION IS CUT

An activated installation keeps operating with no outside connection at all: instrument control and data recording are local. It only does without the lookups listed above.

CHECK IT YOURSELF

- Run `arcsecond doctor` on the machine: it tests every destination above and sends nothing about your observatory. `arcsecond doctor --report` writes the result to a file for your records.
- This sheet is generated from the software's own network manifest (audited 2026-10-05); the current version is at docs.arcsecond.io/reference/network.
- Security questions: team@arcsecond.io